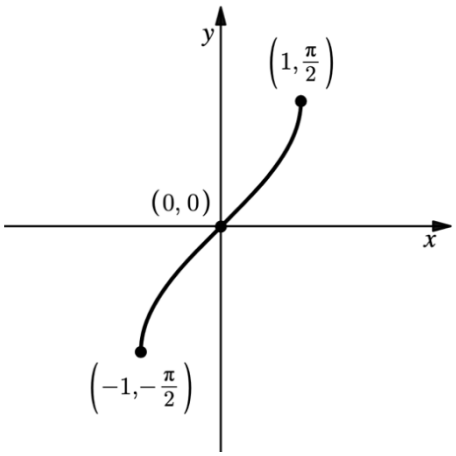
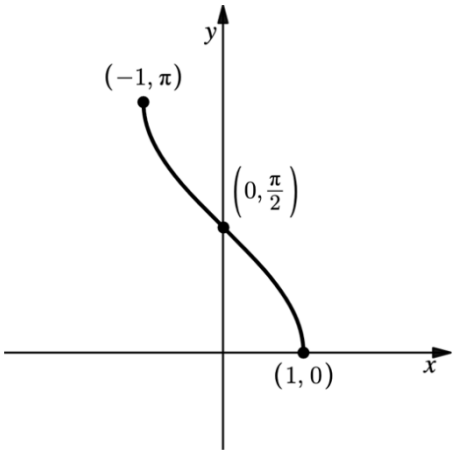
2.14 Trigonometry: Inverse Trigonometric Functions

Worked Solutions 1–27

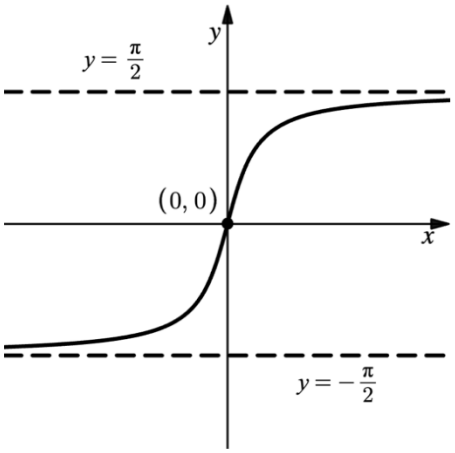
1 a)



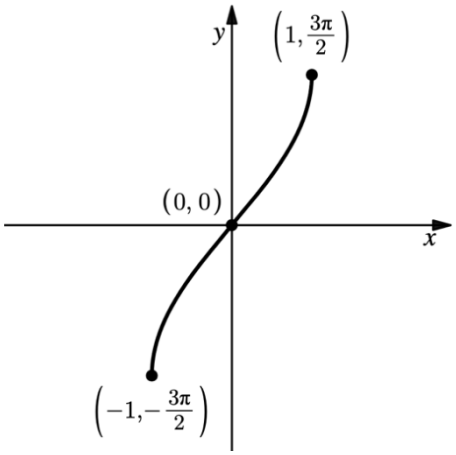
1 b)



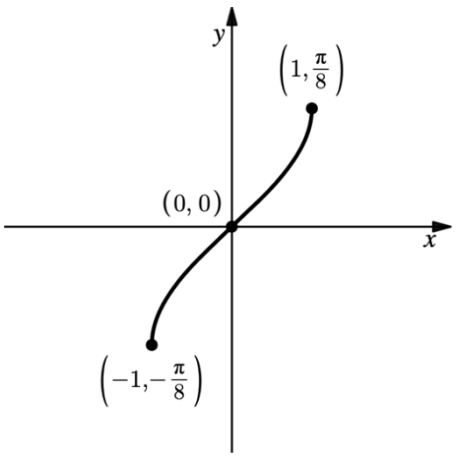
1 c)


y
$y = \frac{\pi}{2}$
$(0, 0)$
x
$y = -\frac{\pi}{2}$

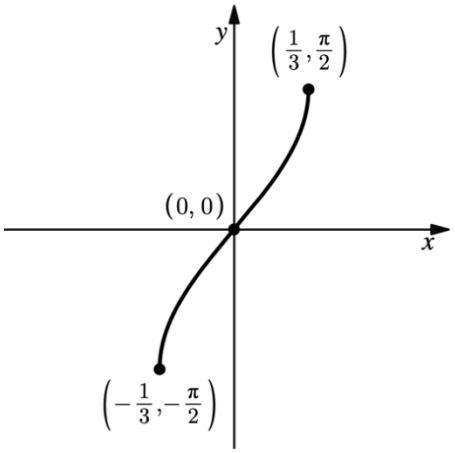
2 a)


y
$\left(1, \frac{3\pi}{2}\right)$
$(0, 0)$
x
$\left(-1, -\frac{3\pi}{2}\right)$

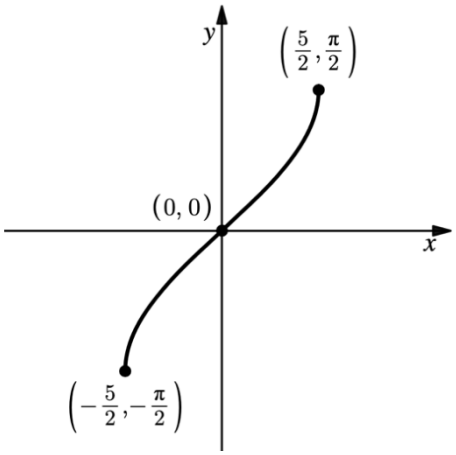
2 b)


y
$\left(1, \frac{\pi}{8}\right)$
$(0, 0)$
x
$\left(-1, -\frac{\pi}{8}\right)$

3 a)



3 b)



4 a)

$y = \arcsin x + \frac{\pi}{2}$.

4 b)

$y = -\arcsin x$ or $y = \arcsin(-x)$.

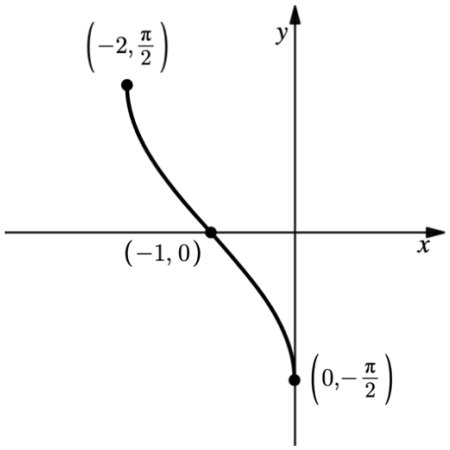
4 c)

$y = \arcsin(x - 1)$.

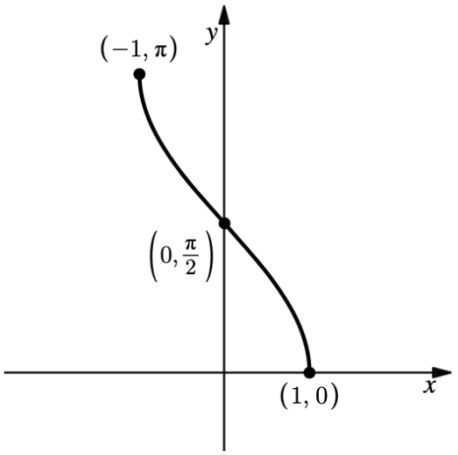
4 d)

$y = \arcsin(x + 1) - \frac{\pi}{2}$.

5)



6)



7)

$$\pi - 3\arcsin(2x-3) = 0 \Rightarrow \arcsin(2x-3) = \frac{\pi}{3},$$
$$2x - 3 = \sin\frac{\pi}{3} \Rightarrow 2x - 3 = \frac{\sqrt{3}}{2} \Rightarrow x = \frac{\frac{\sqrt{3}}{2} + 3}{2},$$
$$x = \frac{\sqrt{3}}{4} + \frac{3}{2}.$$

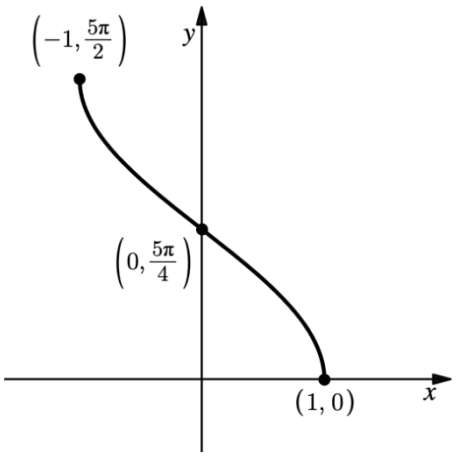
8 a)

$f(x) = 2 - 3\sin x$. Let $y = f(x) \Rightarrow y = 2 - 3\sin x$. Swap xs and ys: $x = 2 - 3\sin y$. Rearrange to make y the subject: $x = 2 - 3\sin y \Rightarrow 3\sin y = 2 - x \Rightarrow \sin y = \frac{2-x}{3} \Rightarrow y = \arcsin\left(\frac{2-x}{3}\right)$.

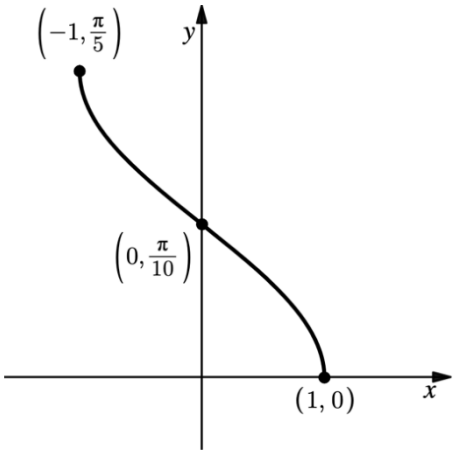
8 b)

Domain of $f^{-1}(x)$: $\{x : x \in \mathbb{R}, -1 \leq x \leq 5\}$. Range of $f^{-1}(x)$: $\{f^{-1}(x) : f^{-1}(x) \in \mathbb{R}, -\frac{\pi}{2} \leq f^{-1}(x) \leq \frac{\pi}{2}\}$.

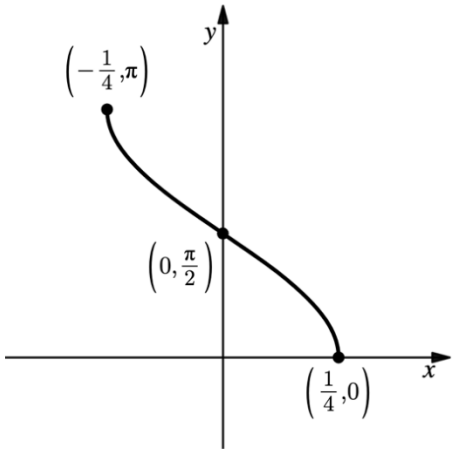
9 a)



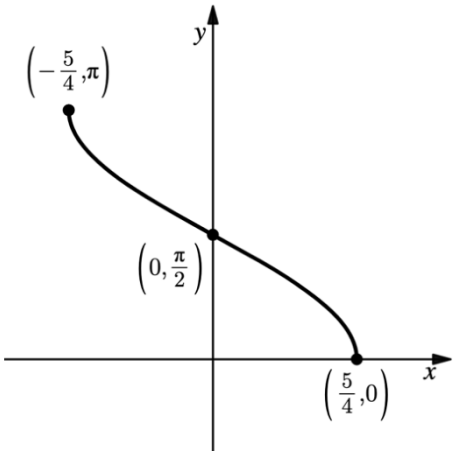
9 b)



10 a)



10 b)



11 a)

$y = -\arccos x$.

11 b)

$y = \arccos(-x)$.

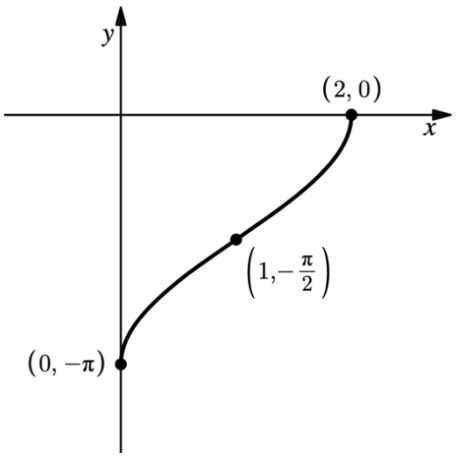
11 c)

$y = \arccos x - \frac{\pi}{2}$.

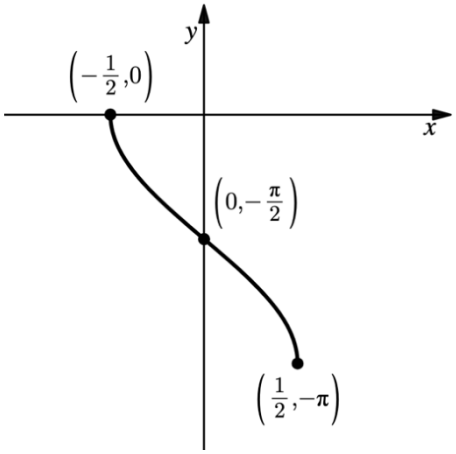
11 d)

$y = \arccos(x+1) - \pi$.

12)



13)



14)

$$6\pi - 9\arccos(\tfrac{1}{5}x+3) = 0 \Rightarrow \arccos(\tfrac{1}{5}x+3) = \frac{2\pi}{3},$$
$$\tfrac{1}{5}x+3 = \cos\frac{2\pi}{3} \Rightarrow \tfrac{1}{5}x+3 = -\frac{1}{2} \Rightarrow \tfrac{1}{5}x = -\frac{7}{2} \Rightarrow x = -\frac{35}{2}.$$

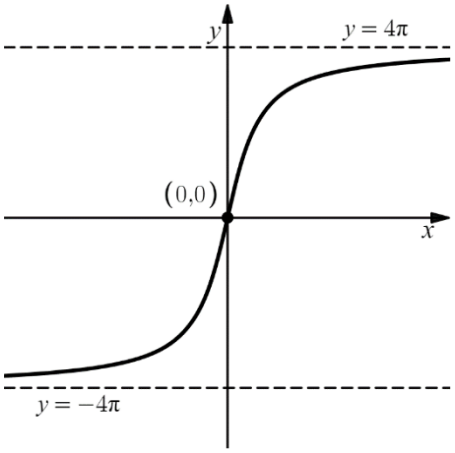
15 a)

$\arccos x = \frac{\pi}{3} \Rightarrow x = \cos\frac{\pi}{3} = \frac{1}{2}$.

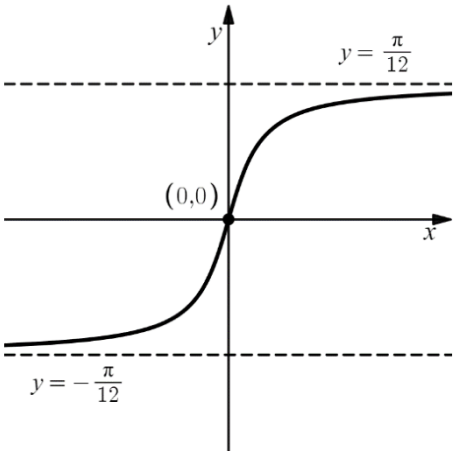
15 b)

$\arcsin x = \theta \Rightarrow \sin\theta = x.\ \sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$.

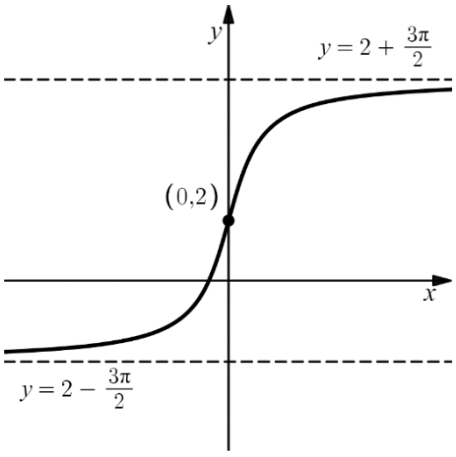
16 a)


y
$y = 4\pi$
(0,0)
x
$y = -4\pi$

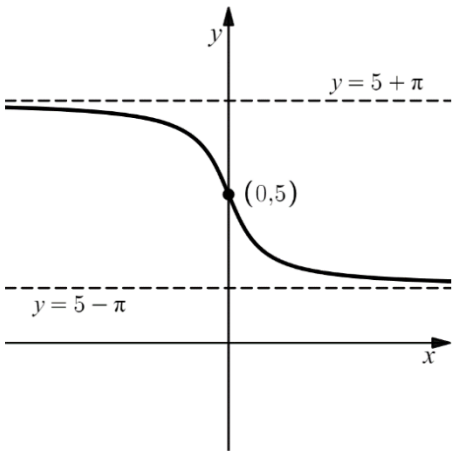
16 b)


y
$y = \frac{\pi}{12}$
(0,0)
x
$y = -\frac{\pi}{12}$

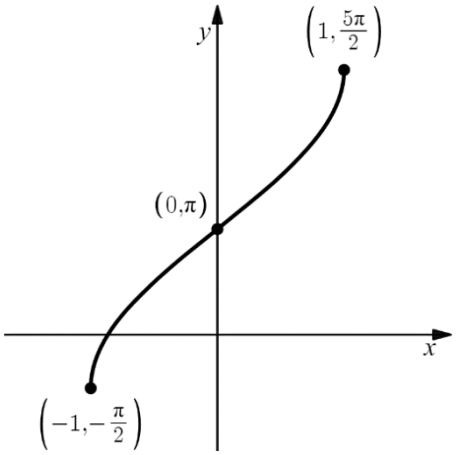
17 a)


y
$y = 2 + \frac{3\pi}{2}$
(0,2)
x
$y = 2 - \frac{3\pi}{2}$

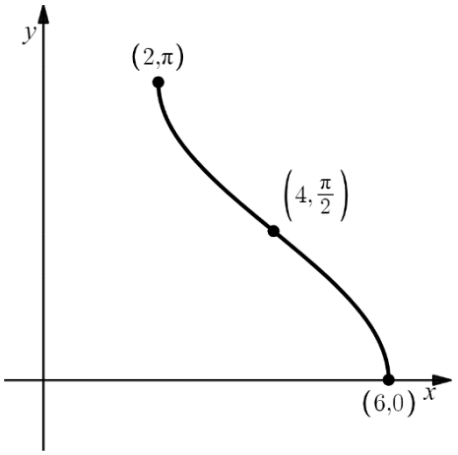
17 b)


y
$y = 5 + \pi$
$(0,5)$
$y = 5 - \pi$
x

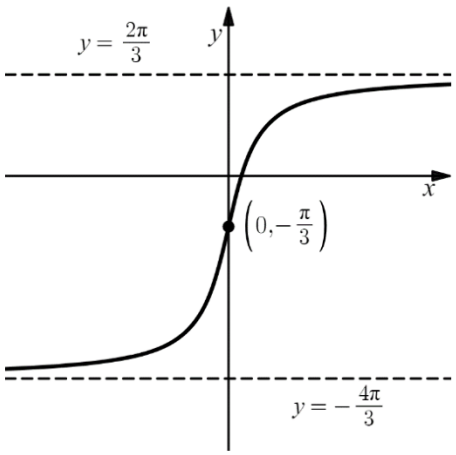
18 a)


y
$\left(1, \frac{5\pi}{2}\right)$
$(0,\pi)$
x
$\left(-1, -\frac{\pi}{2}\right)$

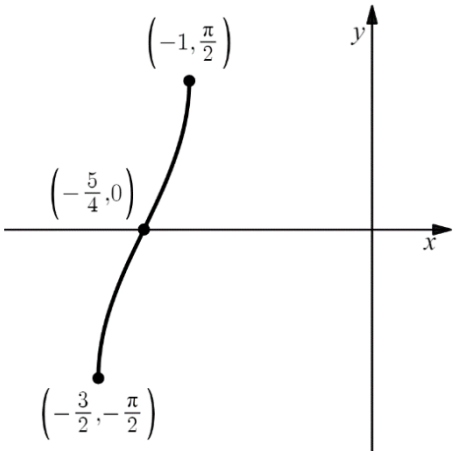
18 b)


y
$(2,\pi)$
$\left(4, \frac{\pi}{2}\right)$
$(6,0)$
x

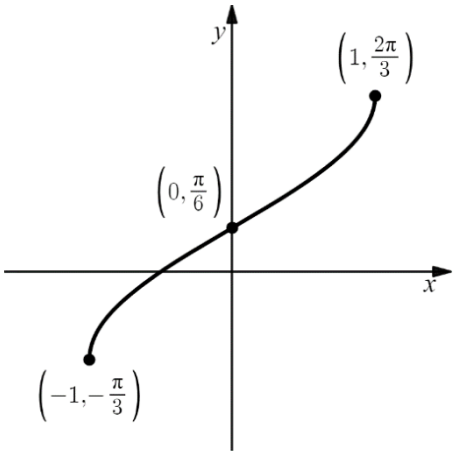
18 c)


$y = \frac{2\pi}{3}$
y
x
$\left(0, -\frac{\pi}{3}\right)$
$y = -\frac{4\pi}{3}$

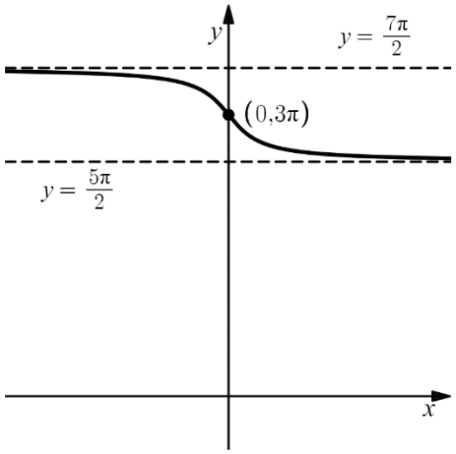
18 d)


$\left(-1, \frac{\pi}{2}\right)$
y
$\left(-\frac{5}{4}, 0\right)$
x
$\left(-\frac{3}{2}, -\frac{\pi}{2}\right)$

18 e)


y
$\left(1, \frac{2\pi}{3}\right)$
$\left(0, \frac{\pi}{6}\right)$
x
$\left(-1, -\frac{\pi}{3}\right)$

18 f)



19 a)

$y = 2\arcsin x + \frac{\pi}{4}$.

$y = \frac{5\pi}{4} - 2\arccos x$.

19 b)

$y = \frac{\pi}{2} + \arcsin(\frac{1}{4} - 2x)$.

$y = \arccos(2x - \frac{1}{4})$.

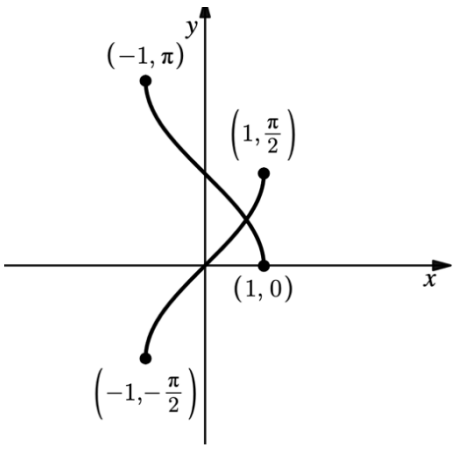
19 c)

$y = \arcsin(3x + \frac{5}{6})$.

$y = \frac{\pi}{2} - \arccos(3x + \frac{5}{6})$.

20)

Let $y = \arcsin x$ then $y = \arccos x$ also. $\sin y = x$ and $\cos y = x$. We know that $\sin^2 y + \cos^2 y = 1 \Rightarrow x^2 + x^2 = 1 \Rightarrow 2x^2 = 1 \Rightarrow x^2 = \frac{1}{2} \Rightarrow x = \pm\frac{1}{\sqrt{2}}$.

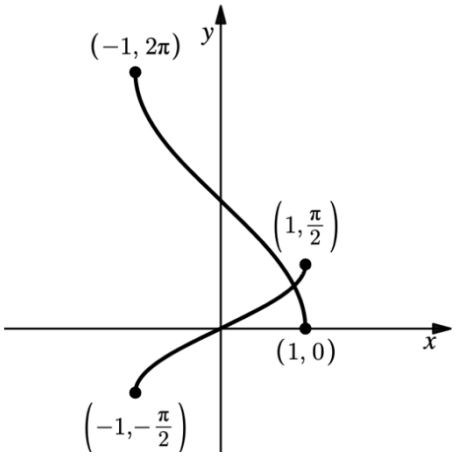


As can be seen from the diagram above, there is only one point of intersection and that is when x is positive, so the only solution is $x = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$.

21 a)

Let $y = \arcsin x$ then $y = 2\arccos x$ also. $\sin y = x$ and $\cos(\frac{1}{2}y) = x$. $\cos 2A = 2\cos^2 A - 1 \Rightarrow \cos y = 2\cos^2(\frac{1}{2}y) - 1 \Rightarrow \cos y = 2x^2 - 1$. We know that $\sin^2 y + \cos^2 y = 1$.

$$x^2 + (2x^2 - 1)^2 = 1 \Rightarrow x^2 + 4x^4 - 4x^2 + 1 = 1,$$
$$4x^4 - 3x^2 = 0 \Rightarrow (4x^2 - 3)x^2 = 0,$$
$$x = 0 \text{ or } 4x^2 - 3 = 0 \Rightarrow x = \pm\frac{\sqrt{3}}{2}.$$

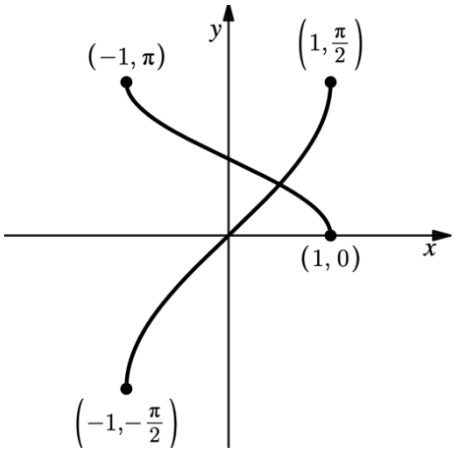


As can be seen from the diagram above, there is only one point of intersection and that is when x is positive, so the only solution is $x = \frac{\sqrt{3}}{2}$.

21 b)

Let $y = 2\arcsin x$ then $y = \arccos x$ also. $\sin(\frac{1}{2}y) = x$ and $\cos y = x$. $\cos 2A = 2\cos^2 A - 1 \Rightarrow \cos y = 2\cos^2(\frac{1}{2}y) - 1$. $x = 2\cos^2(\frac{1}{2}y) - 1 \Rightarrow \cos^2(\frac{1}{2}y) = \frac{x+1}{2} \Rightarrow \cos(\frac{1}{2}y) = \sqrt{\frac{x+1}{2}}$. $\sin 2A = 2\sin A\cos A \Rightarrow \sin y = 2\sin(\frac{1}{2}y)\cos(\frac{1}{2}y) \Rightarrow \sin y = 2x\sqrt{\frac{x+1}{2}}$. We know that $\sin^2 y + \cos^2 y = 1$.

$$\left(2x\sqrt{\frac{x+1}{2}}\right)^2 + x^2 = 1 \Rightarrow 4x^2\left(\frac{x+1}{2}\right) + x^2 = 1,$$
$$2x^3 + 2x^2 + x^2 - 1 = 0 \Rightarrow 2x^3 + 3x^2 - 1 = 0,$$
$$(2x - 1)(x + 1)^2 = 0 \Rightarrow x = \frac{1}{2} \text{ or } x = -1.$$



As can be seen from the diagram above, there is only one point of intersection and that is when x is positive, so the only solution is $x = \frac{1}{2}$.

22)

$\arcsin x = \frac{\pi}{5} \Rightarrow x = \sin\frac{\pi}{5}$. Let $\theta = \frac{\pi}{5} \Rightarrow 5\theta = \pi$. Subtract 2θ from both sides: $3\theta = \pi - 2\theta$. Take sine of both sides: $\sin 3\theta = \sin(\pi - 2\theta)$. We have that $\sin(\pi - A) = \sin A$. $\therefore \sin 3\theta = \sin 2\theta$. Use the triple and double angle formulae:

$$\begin{aligned} 3\sin\theta - 4\sin^3\theta &= 2\sin\theta\cos\theta, \\ (3 - 4\sin^2\theta)\sin\theta &= 2\sin\theta\cos\theta, \\ 3 - 4\sin^2\theta &= 2\cos\theta \text{ as } \sin\theta > 0, \\ 3 - 4(1 - \cos^2\theta) &= 2\cos\theta, \\ 3 - 4 + 4\cos^2\theta &= 2\cos\theta, \\ 4\cos^2\theta - 2\cos\theta - 1 = 0 &\Rightarrow \cos\theta = \frac{1 \pm \sqrt{5}}{4}. \end{aligned}$$

So $\cos\theta = \frac{1+\sqrt{5}}{4}$ as $\cos\theta > 0$.

$$\sin^2\theta + \cos^2\theta = 1 \Rightarrow \sin^2\theta + \left(\frac{1+\sqrt{5}}{4}\right)^2 = 1,$$

$$\sin\theta = \sqrt{1 - \left(\frac{1+\sqrt{5}}{4}\right)^2} \Rightarrow \sin\theta = \sqrt{1 - \frac{3+\sqrt{5}}{8}},$$

$$\sin\theta = \sqrt{\frac{5-\sqrt{5}}{8}}.$$

So $\sin\frac{\pi}{5} = \sqrt{\frac{5-\sqrt{5}}{8}} \Rightarrow \arcsin\sqrt{\frac{5-\sqrt{5}}{8}} = \frac{\pi}{5}$.

23 a)

$\arcsin 1 = \frac{\pi}{2}$.

23 b)

$\arccos(-1) = \pi$.

23 c)

$\arcsin 0 = 0$.

23 d)

$\arccos 1 = 0$.

24 a)

$\arcsin\frac{\sqrt{2}}{2} = \frac{\pi}{4}$.

24 b)

$\arccos\frac{\sqrt{2}}{2} = \frac{\pi}{4}$.

24 c)

$\arcsin\left(-\frac{\sqrt{2}}{2}\right) = -\frac{\pi}{4}$.

24 d)

$\arccos\left(-\frac{\sqrt{2}}{2}\right) = \frac{3\pi}{4}$.

25 a)

$\arctan 1 = \frac{\pi}{4}$.

25 b)

$\arctan(-1) = -\frac{\pi}{4}$.

25 c)

$\arctan\sqrt{3} = \frac{\pi}{3}$.

25 d)

$\arctan\left(-\frac{\sqrt{3}}{3}\right) = -\frac{\pi}{6}$.

26 a)

$\arcsin\left(\sin\frac{\pi}{2}\right) = \frac{\pi}{2}$.

26 b)

$\arccos\left(\cos\left(-\frac{\pi}{2}\right)\right) = \frac{\pi}{2}$.

26 c)

$\arcsin\left(\sin\frac{3\pi}{2}\right) = -\frac{\pi}{2}$.

26 d)

$\arccos\left(\cos\left(-\frac{3\pi}{2}\right)\right) = \frac{\pi}{2}$.

26 e)

$\arcsin\left(\sin\frac{\pi}{3}\right) = \frac{\pi}{3}$.

26 f)

$\arccos\left(\cos\frac{2\pi}{3}\right) = \frac{2\pi}{3}$.

26 g)

$\arcsin\left(\sin\frac{5\pi}{3}\right) = -\frac{\pi}{3}$.

26 h)

$\arccos\left(\cos\left(-\frac{\pi}{3}\right)\right) = \frac{\pi}{3}$.

26 i)

$\arctan\left(\tan\frac{\pi}{4}\right) = \frac{\pi}{4}$.

26 j)

$\arctan\left(\tan\frac{2\pi}{3}\right) = -\frac{\pi}{3}$.

27 a)

$\sin\left(\arcsin\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}}{2}$.

27 b)

$\cos\left(\arccos\left(-\frac{\sqrt{2}}{2}\right)\right) = -\frac{\sqrt{2}}{2}$.

27 c)

$\sin\left(\arccos\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$.

27 d)

$\cos\left(\arcsin\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$.

27 e)

$\sin\left(\arccos\frac{\sqrt{3}}{2}\right) = \frac{1}{2}$.

27 f)

$\cos\left(\arcsin\left(-\frac{\sqrt{3}}{2}\right)\right) = \frac{1}{2}$.

27 g)

$\tan\left(\arcsin\left(-\frac{1}{2}\right)\right) = -\frac{\sqrt{3}}{3}$.